

SYNTHETIC GPR REBAR INVERSION

Recovering rebar geometry from radar waveforms

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A mature FDTD / full-waveform-inversion pipeline that recovers a 3-rebar variable-depth, variable-radius scene under noise, source mismatch and ringdown stress — **with confidence reporting.**

425/425

exact geometry recovered

740–1218

experiments presented

synthetic

not yet field-validated

THE PROBLEM

Find hidden rebar from a single radar trace

Ground-penetrating radar sees concrete from the surface. We want each rebar's horizontal position (x), depth (z) and radius (r) — but the only measurement is the reflected waveform.



Forward model: FDTD

Finite-difference time-domain simulation turns any candidate concrete/rebar scene into a synthetic radar waveform.



Inverse model: FWI

The optimizer never sees the truth. It proposes scenes and keeps the one whose simulated waveform best matches the reference.



Three synthetic rebars

target0 (150, 80, r5) · target1 (250, 100, r6) · target2 (350, 120, r8), in millimetres.

THE OPTIMIZER LOOP

1

Propose a candidate scene

guess x, z, r in bounded windows

2

Simulate its waveform

run FDTD forward

3

Score the misfit

candidate vs reference (lower is better)

4

Pick the best, report margin

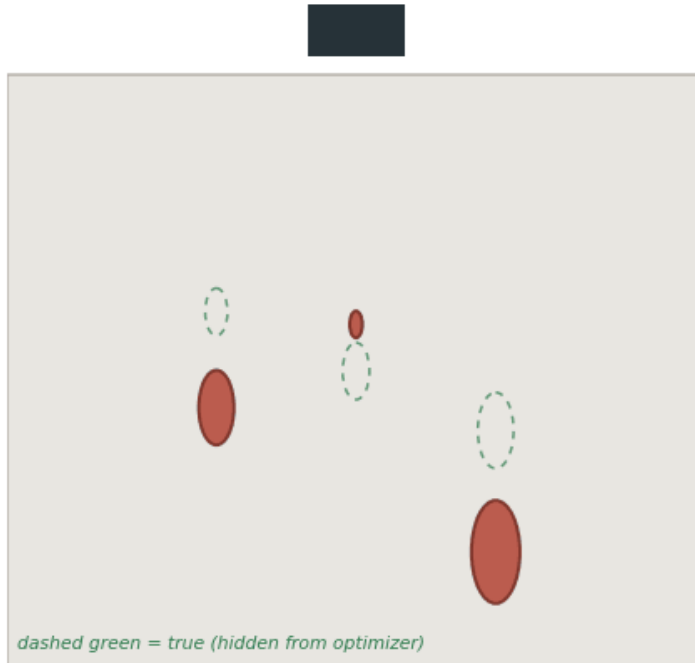
best radius vs runner-up radius

loops until the misfit stops improving

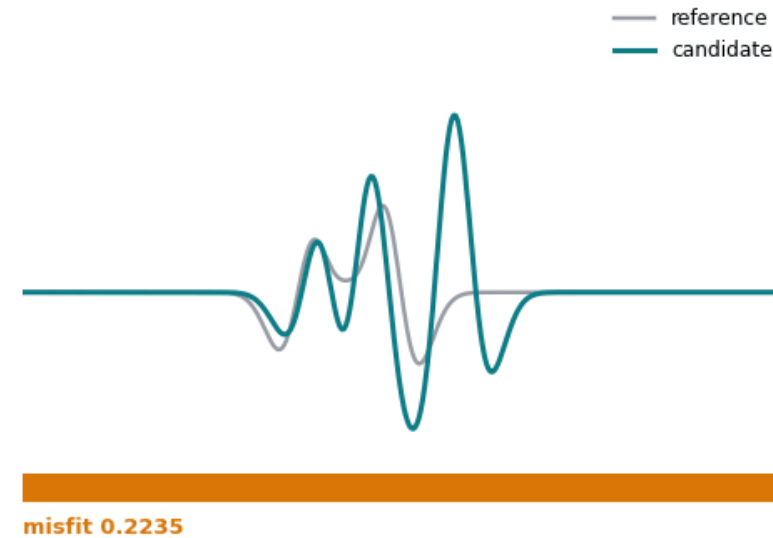
Forward simulation and inverse recovery

Inverse search: the optimizer adjusts x , z , r until the waveforms match

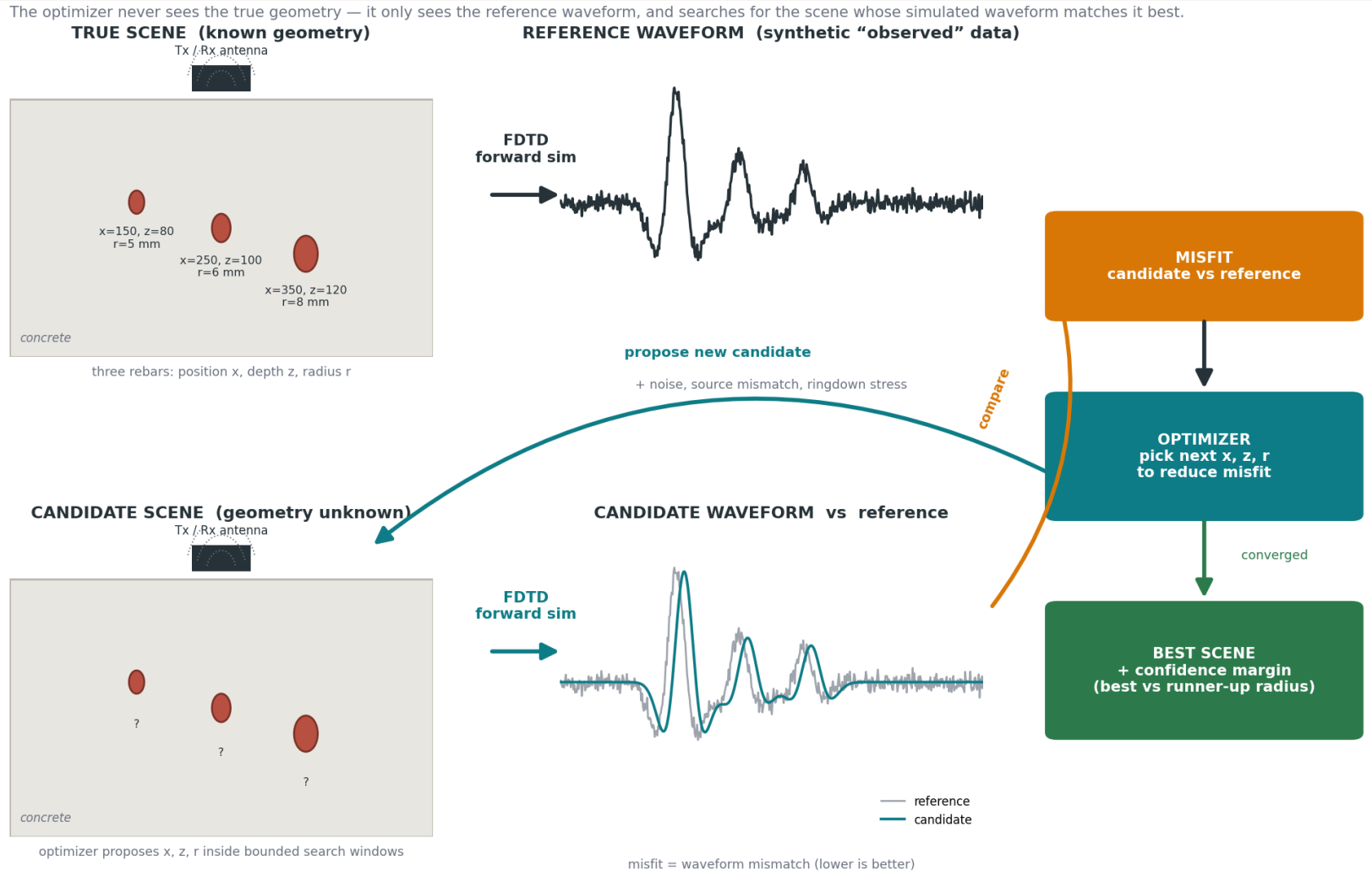
Candidate scene (search step 1/26)



Candidate waveform vs reference



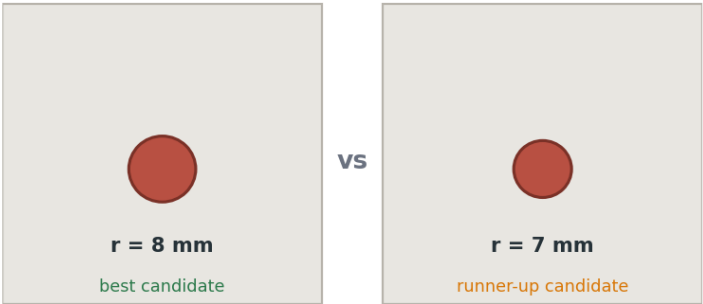
Forward simulation and inverse recovery



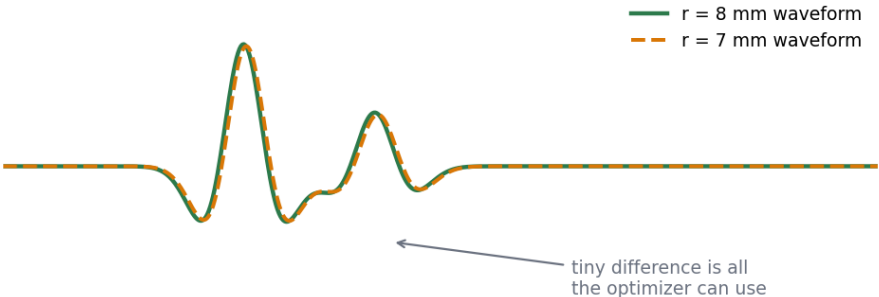
Radius ambiguity & confidence margin

Across runs 740-1218, all 425 runs selected the exact $x/z/r$ — the open problem is how clearly the best radius beats the runner-up.

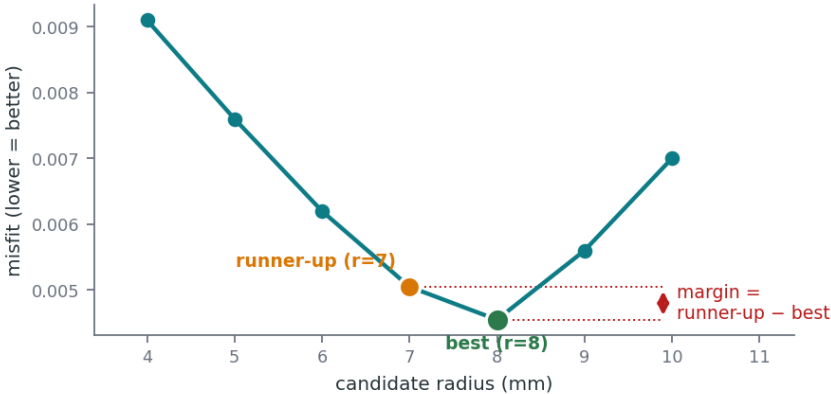
Two competing scenes
(same x, z — only radius differs)



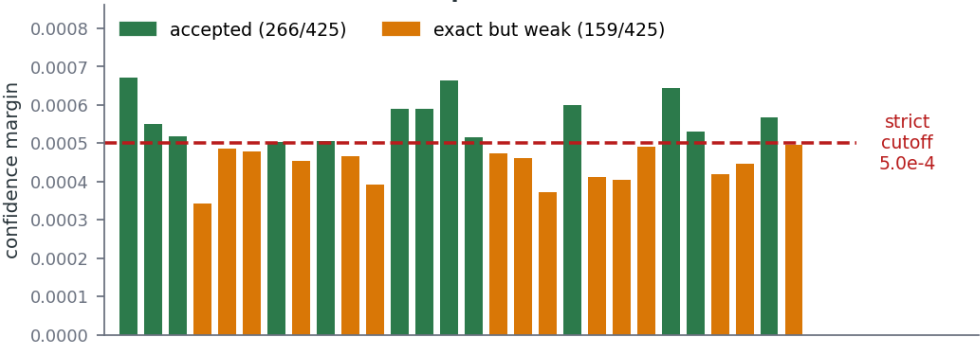
...produce nearly identical waveforms



Misfit across candidate radii



Every run picks the exact geometry — but margins split runs into accepted vs exact-but-weak



Three workflow levels, one mature core



1 • Single-rebar

Detector-seeded local sizing

- Detects one rebar, seeds an $x/z/r$ refinement
- Handles synthetic noise & source mismatch
- Mature for local sizing — not a field claim

MATURE (LOCAL)



2 • Multi-rebar

Coordinate optimizer (the core)

- 3-rebar variable depth & radius
- Bounded $x/z/r$ search vs reference waveform
- Reports geometry, radius, margin, diagnostics

MATURE CORE



3 • Field-data QC

Real GSSI data — import only

- DZT import, metadata audit, sanity checks
- Kept separate from synthetic experiments
- No validated field inversion yet

QC STAGE ONLY

STRONGEST RESULT

Exact geometry every time, across 425 runs

425

coordinate-optimizer runs parsed

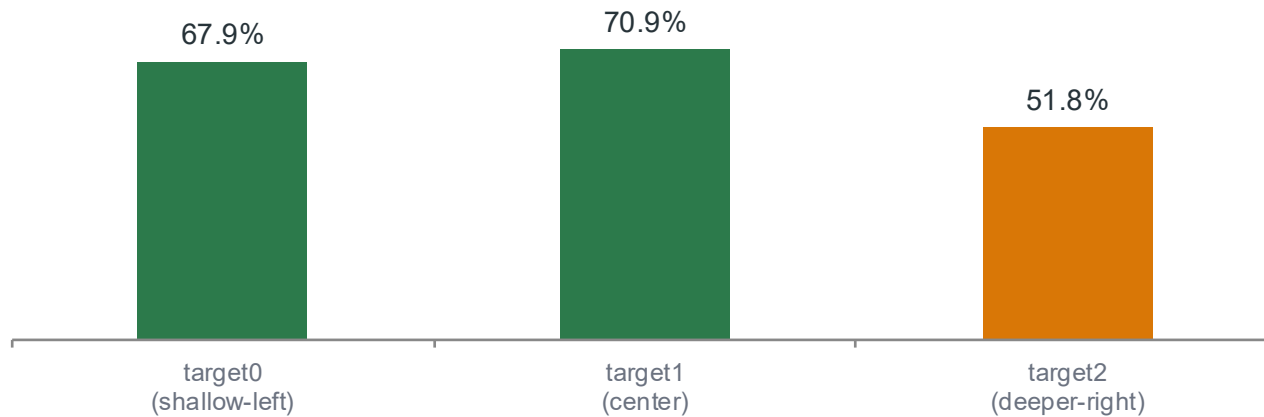
425/425

selected the exact x / z / r

266/425

also cleared the strict cutoff

Acceptance fraction by target



What the numbers mean

Localization is solved.

Every run picked the correct rebar geometry in the tested windows.

target2 is the most fragile

by confidence (51.8% accepted) even though its geometry is still exact.

The issue isn't location — it's confidence

159 of 425 exact-geometry runs stayed below the strict $5.0e-4$ margin. The right rebar is found, but the best radius is sometimes only slightly better than the next competing radius.

Exact-geometry runs



Why it happens

Two scenes that differ only slightly in radius can produce almost identical waveforms. The optimizer must separate them on a tiny misfit difference.



Why it's good practice

Confidence reporting flags the 159 weak rows instead of overstating certainty — it distinguishes exact-and-clear from exact-but-ambiguous.



Where it bites most

target2 (deeper, larger) is the most acceptance-fragile; target1 has exact-but-weak branches that resist simple escalation.

What rescues a weak run — and what doesn't

117 multi-run series tested three intervention families. The lesson: rescues are real but not monotonic — more is not always better.



Source-density

62 series

Often helps target2, but more sources can also worsen the margin.



Tx / Rx offset

33 series

Bracketing rescues acquisition-sensitive cases, especially target0.



Ringdown tuning

22 series

Stable when already in the right regime — not a universal rescue.

CONCRETE EXAMPLE · target2 source-density rescue (runs 1207–1209)

5 sources

margin $4.24\text{e-}4$ **weak**



7 sources

margin $4.94\text{e-}4$ **near cutoff**



9 sources

margin $5.60\text{e-}4$ **accepted**

Confirmed, partial, and still open



CONFIRMED

- Synthetic FDTD forward modeling
- Single-rebar local refinement
- 3-rebar bounded coordinate optimization
- Meaningful confidence reporting
- Objective-variant diagnostics



PARTIAL

- Source-density rescue policies
- Tx/Rx-offset rescue policies
- Ringdown050 replication near cutoff
- target1 exact-but-weak handling



UNRESOLVED

- Real GSSI field-data inversion
- 3D FDTD / FWI & 3D acquisition
- Universal global optimizer
- Synthetic → field threshold transfer
- Coverage of older special cases

RECOMMENDED NEXT STEP

A narrow decision pass — not another long marathon

The project is at a genuine stop-and-evaluate point. The next GPU run should answer one specific policy question, not broadly continue replication.



target1 weak branches

Decide when to use Tx/Rx bracketing instead of adding sources.



target2 near-misses

Decide when to escalate 5 → 7 → 9 source density.



Two cutoffs

Separate strict research confidence from practical reporting confidence.



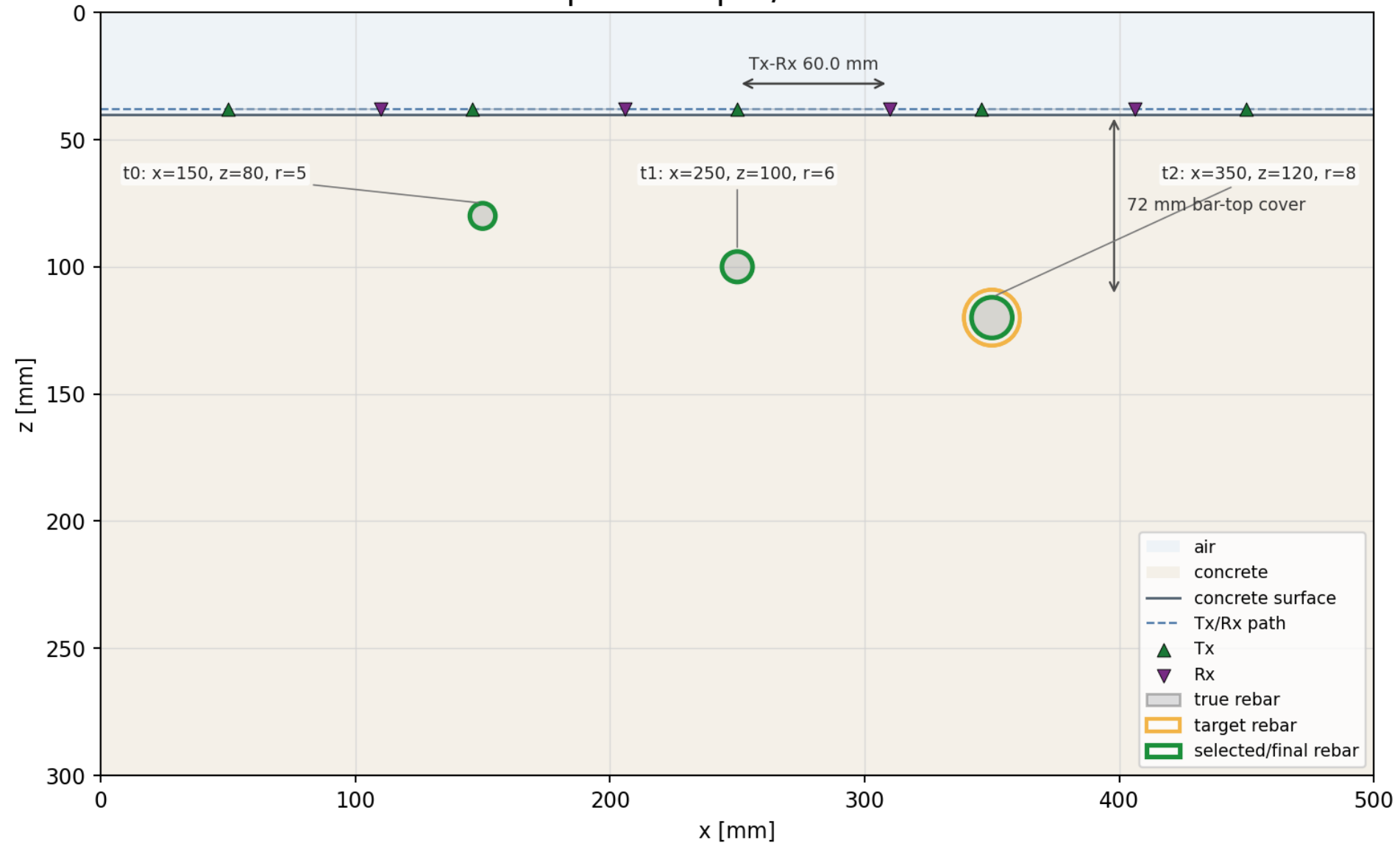
Keep field data apart

Field-data experiments and trackers stay separate from synthetic.

Strongest result today: a mature synthetic inversion-and-reporting pipeline — not yet field-validated, not yet 3D.

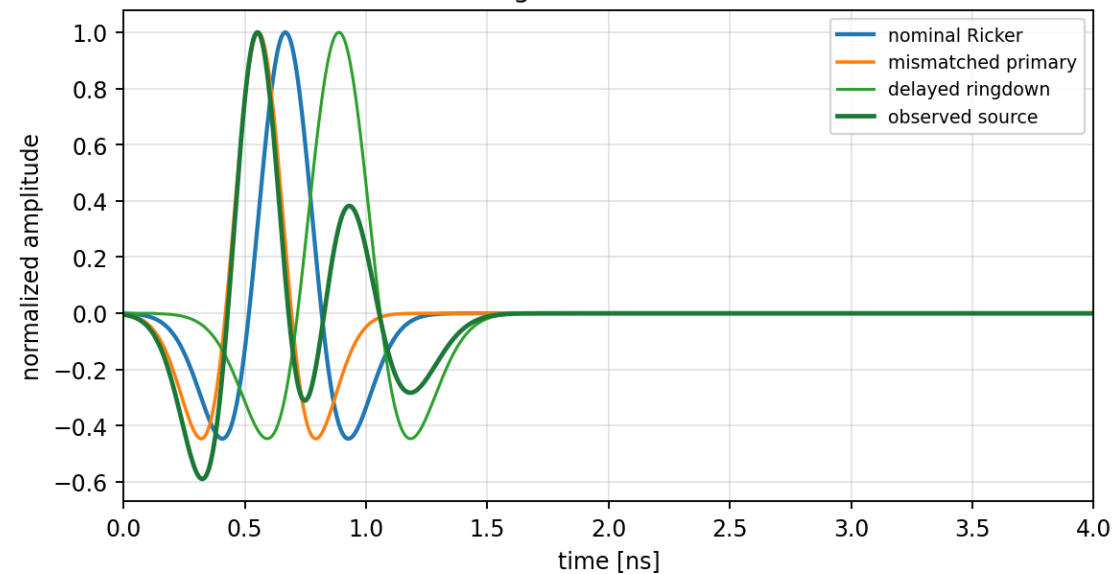
coordinate_optimizer_variable_depth_radius_seed3524578_target2_sources5_txrx60_ringdown050_objectives

5 sources | 1.5 GHz | Tx/Rx offset 60.0 mm

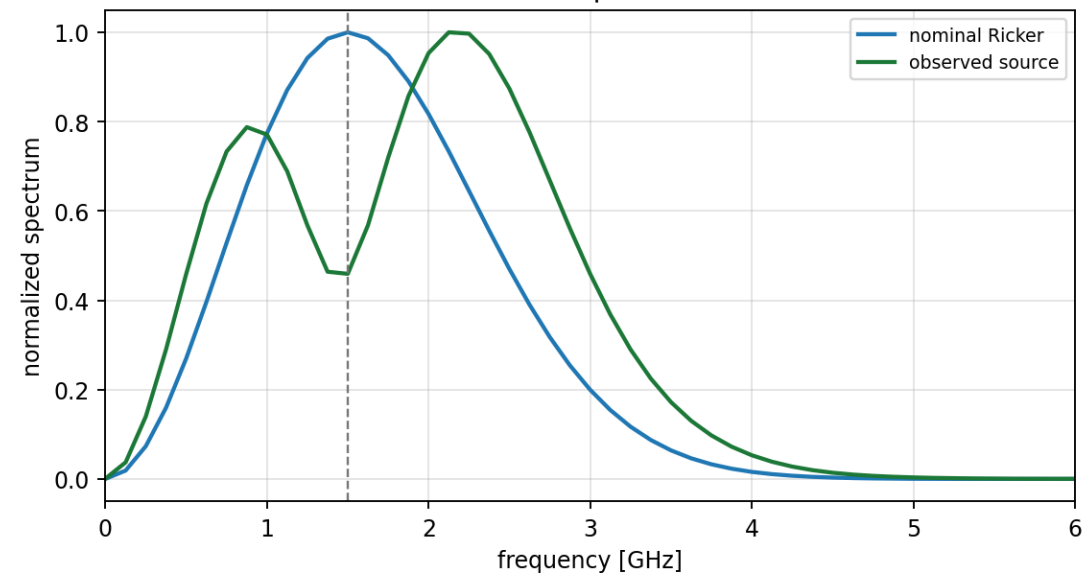


coordinate_optimizer_variable_depth_radius_seed3524578_target2_sources5_txrx60_ringdown050_obj actives

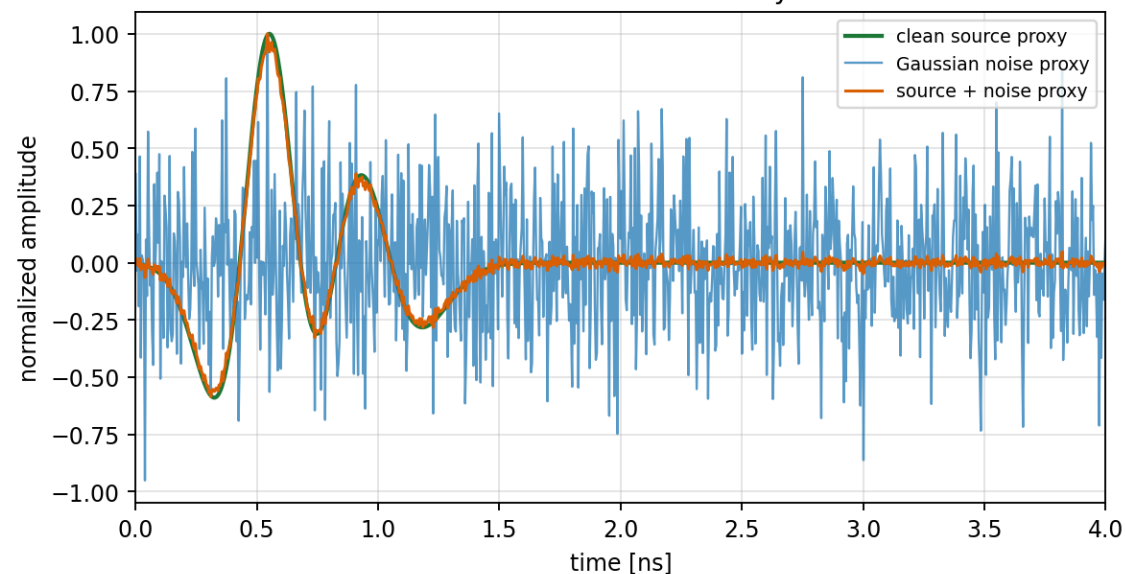
Configured Source Pulse



Source Spectrum



Noise Visualization Proxy



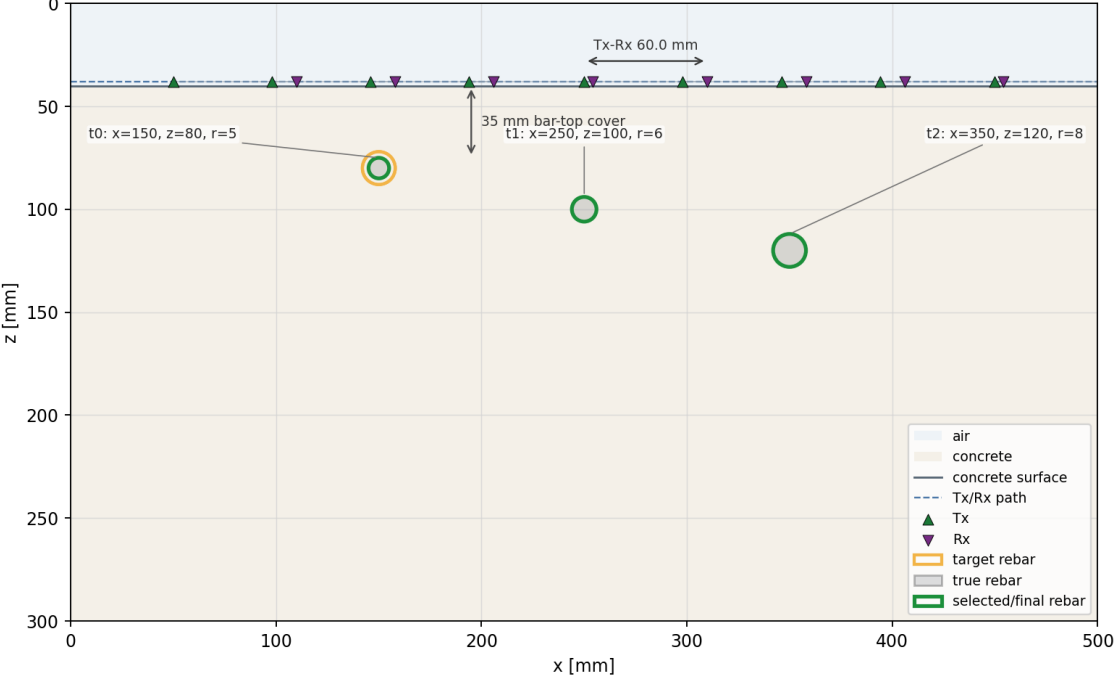
```
case: source mismatch ringdown050_noise10_seed3524578
pulse type: Ricker / Mexican hat
base center frequency: 1.5 GHz
frequency scale: 1.1
time shift: -50 ps
amplitude scale: 1.1
ringdown scale: 0.5
ringdown delay: 180 ps
ringdown frequency scale: 0.8
noise type: zero-mean Gaussian additive observed-data noise
noise RMS fraction: 0.1
noise seed: 3524578
```

observed B-scan noise stats from summary:

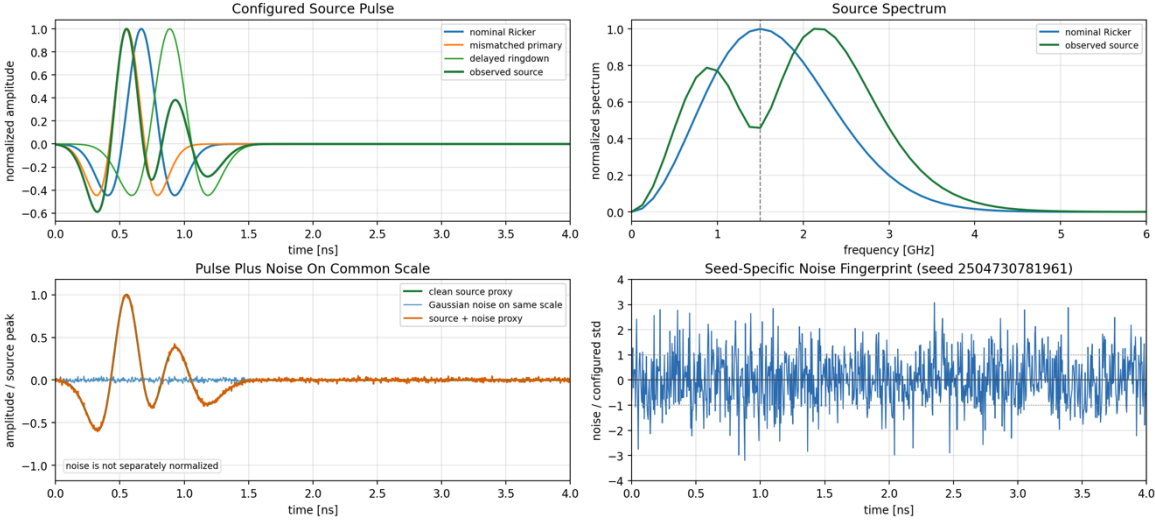
```
clean RMS: 0.000517927
noise std: 5.17927e-05
actual noise RMS: 5.22045e-05
```

Note: the Gaussian noise is added to simulated observed B-scans.
The bottom-left trace is a normalized 1D proxy to show scale and seed.

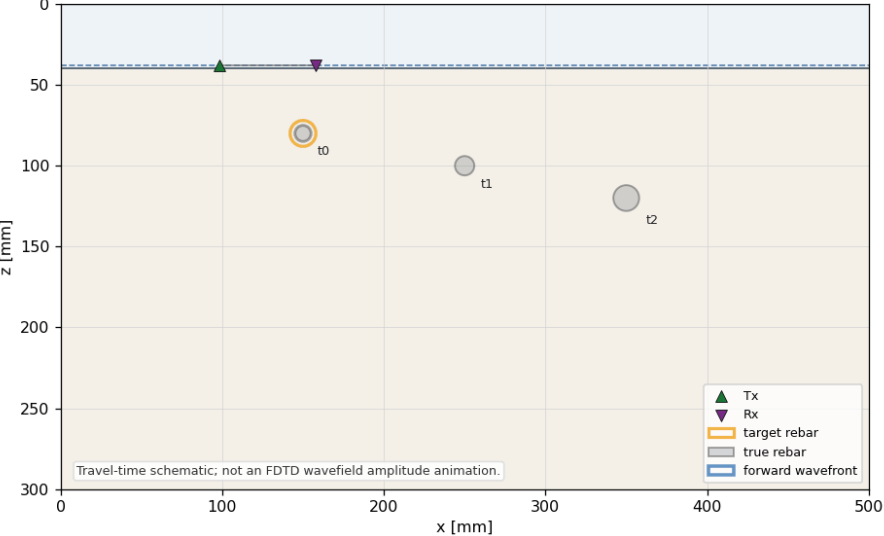
coordinate_optimizer_variable_depth_radius_seed2504730781961_target0_sources9_txrx60_ringdown
n050_objectives scene
9 sources | 1.5 GHz | Tx/Rx offset 60.0 mm



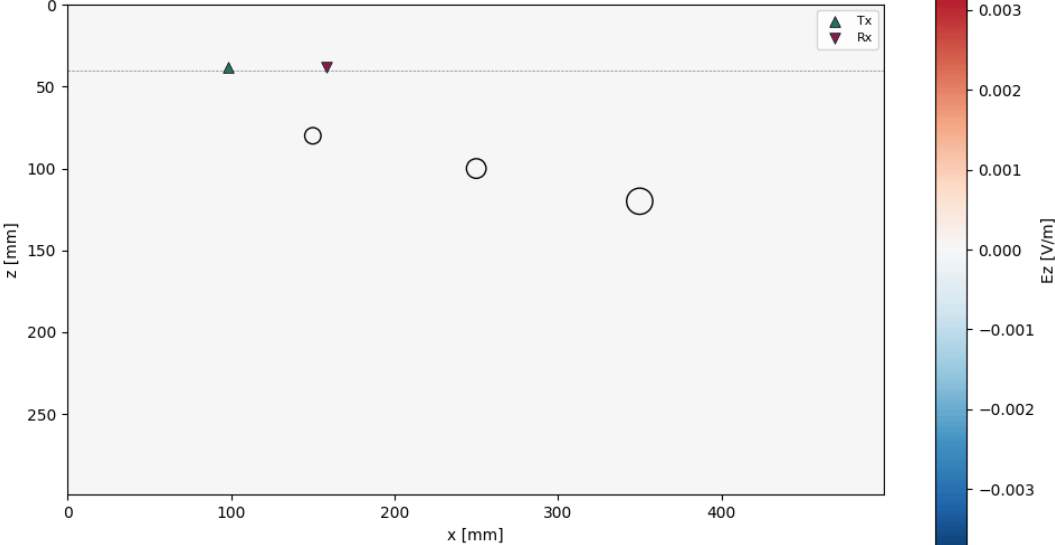
coordinate_optimizer_variable_depth_radius_seed2504730781961_target0_sources9_txrx60_ringdown0
50_objectives

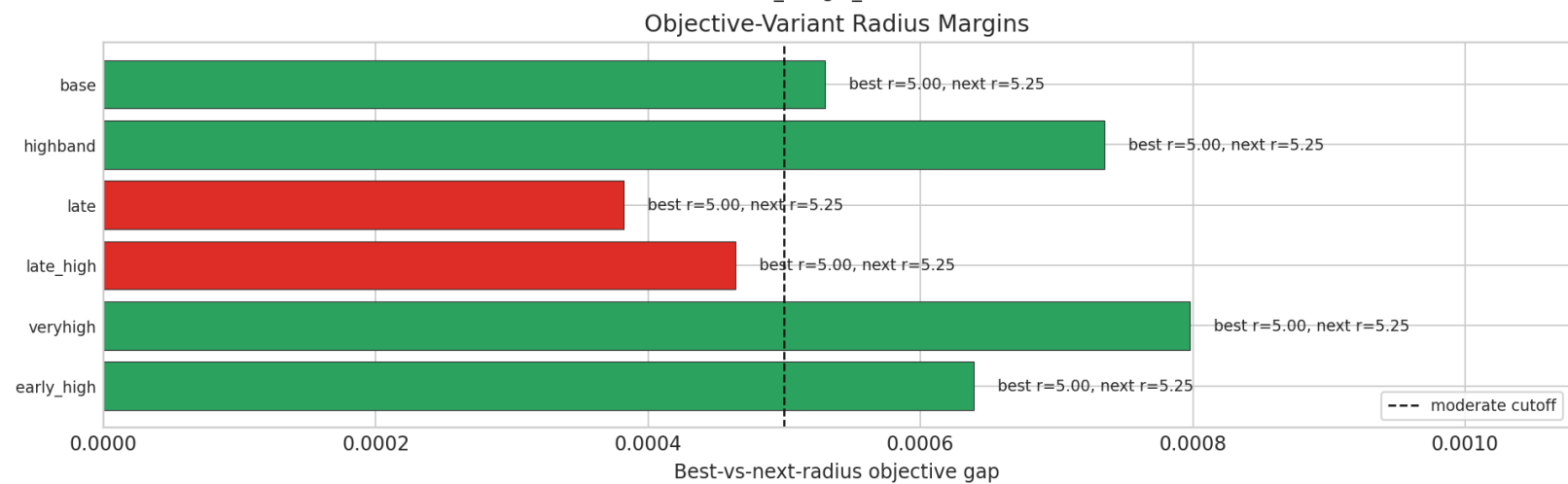
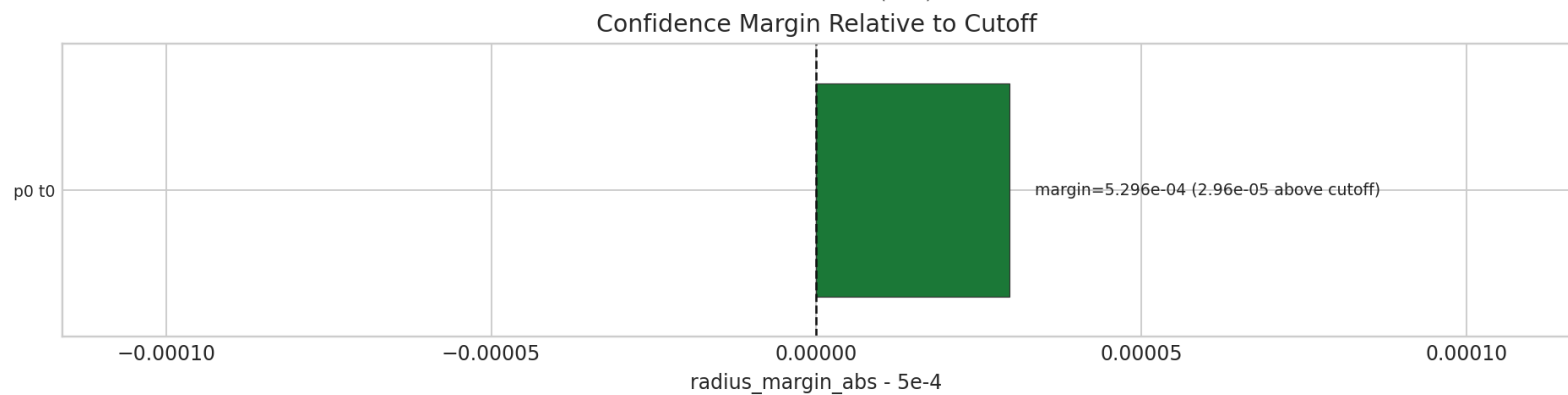
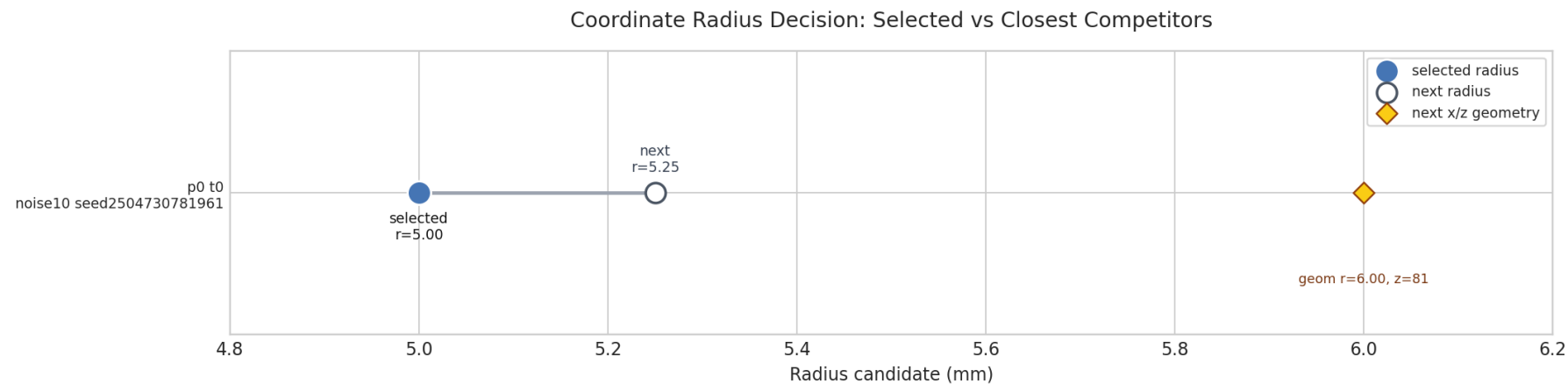


coordinate_optimizer_variable_depth_radius_seed2504730781961_target0_sou
rces9_txrx60_ringdown050_objectives
schematic time 0.00 ns | blue=forward, orange=reflection

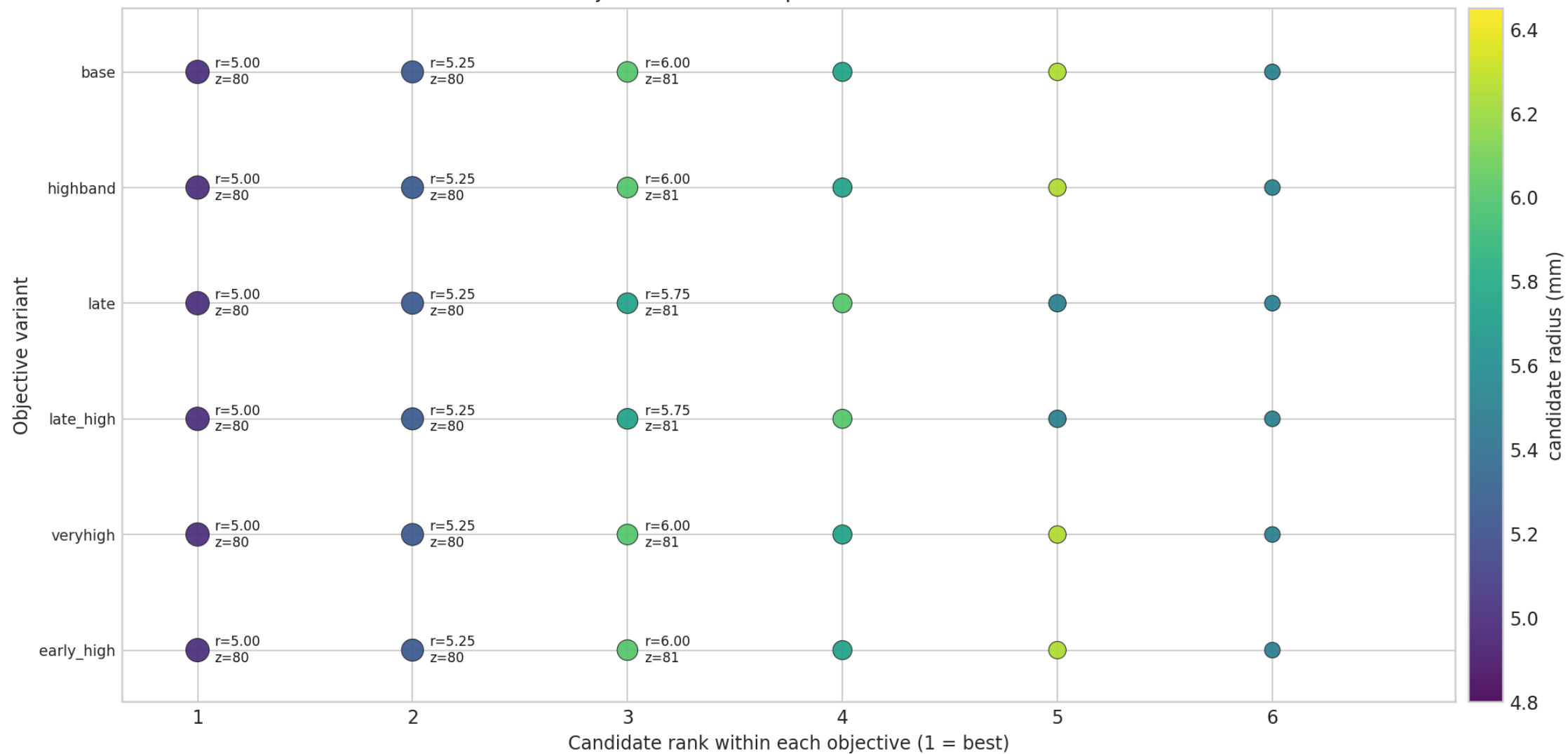


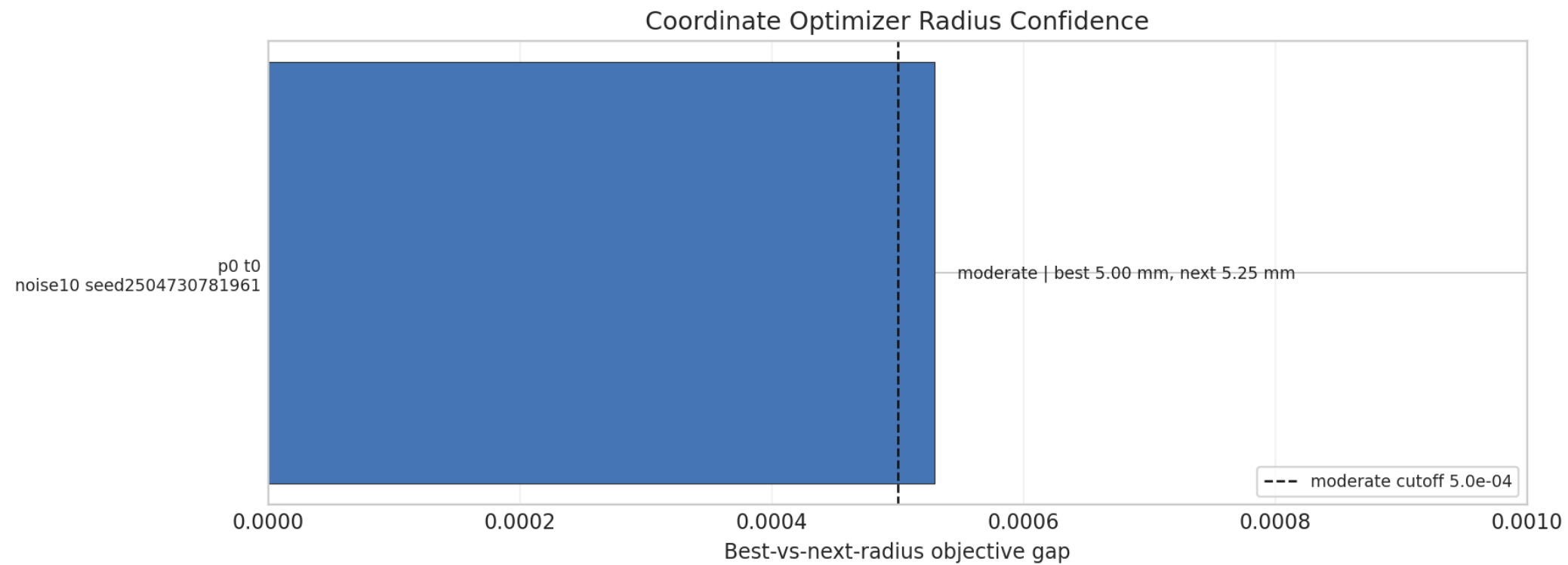
FDTD Ez wavefield | run 1140 target 0 | Tx 98 mm, Rx 158 mm at t = 0.00 ns (step 0)



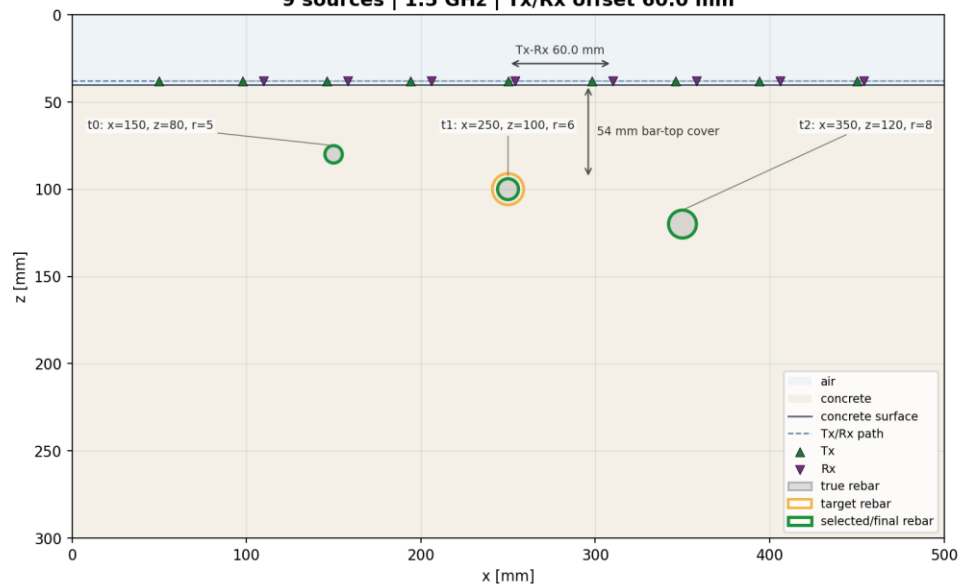


Objective-Variant Top Radius Candidates

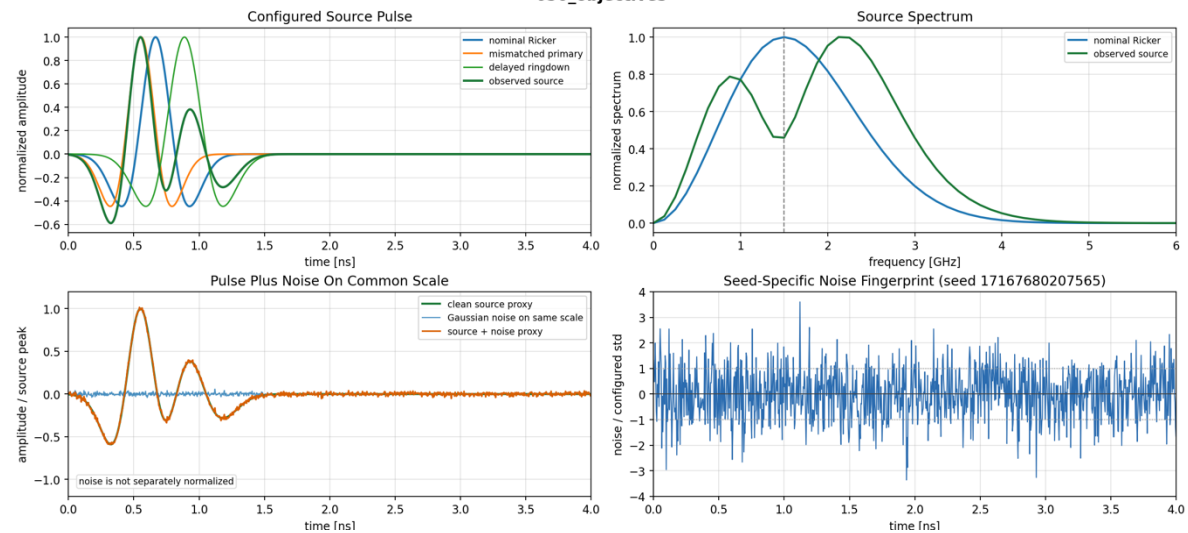




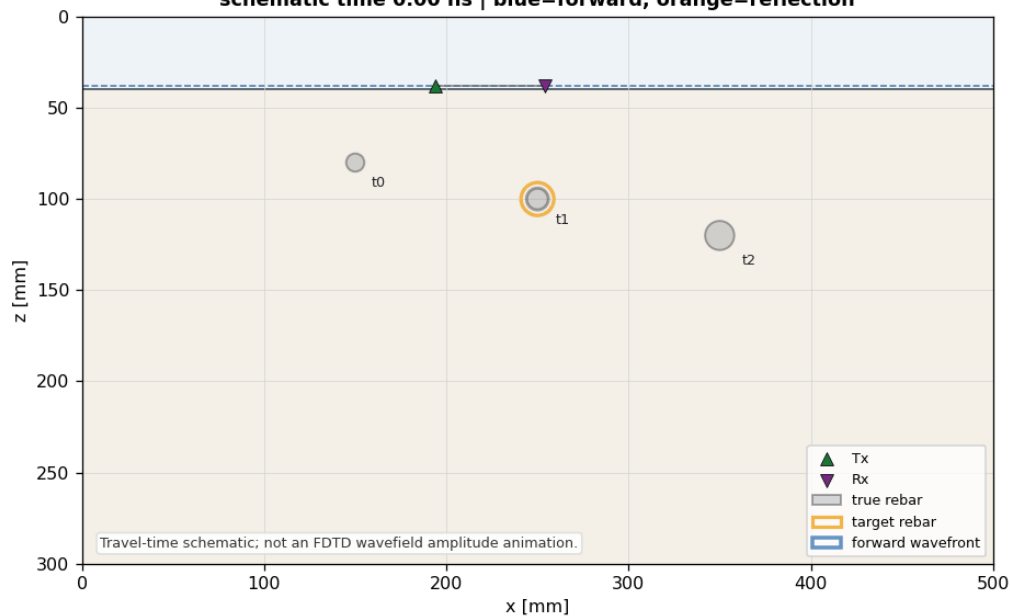
coordinate_optimizer_variable_depth_radius_seed17167680207565_target1_sources9_txrx60_ringdown050_objectives scene
9 sources | 1.5 GHz | Tx/Rx offset 60.0 mm



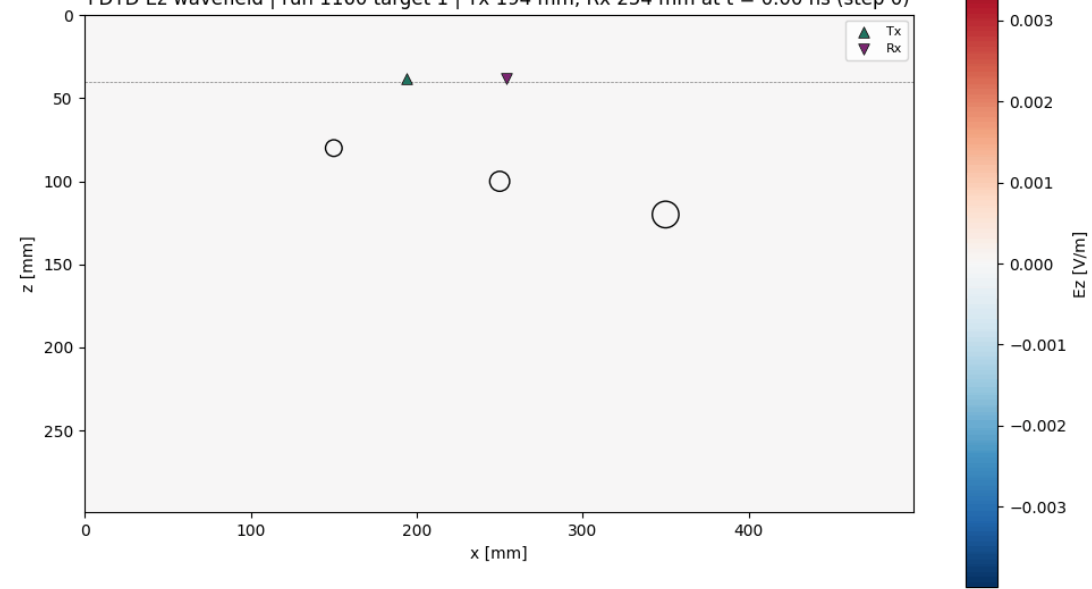
coordinate_optimizer_variable_depth_radius_seed17167680207565_target1_sources9_txrx60_ringdown050_objectives

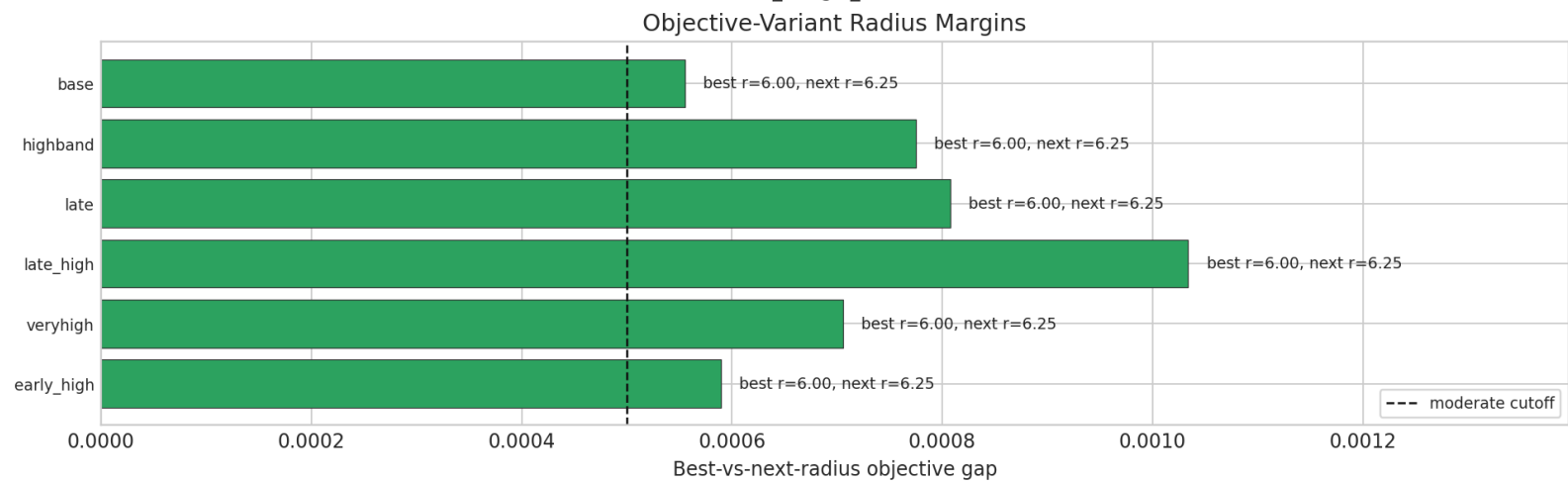
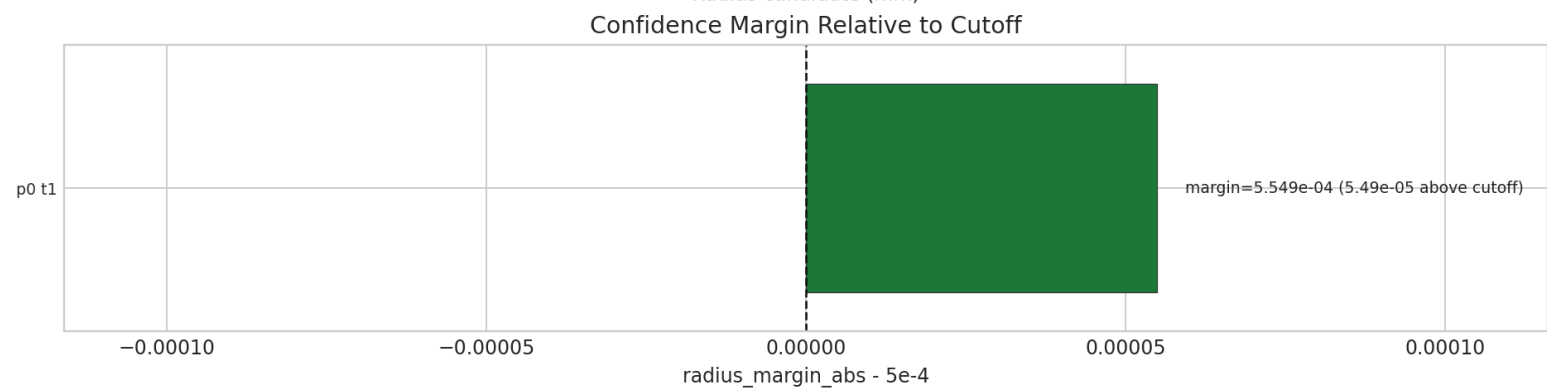
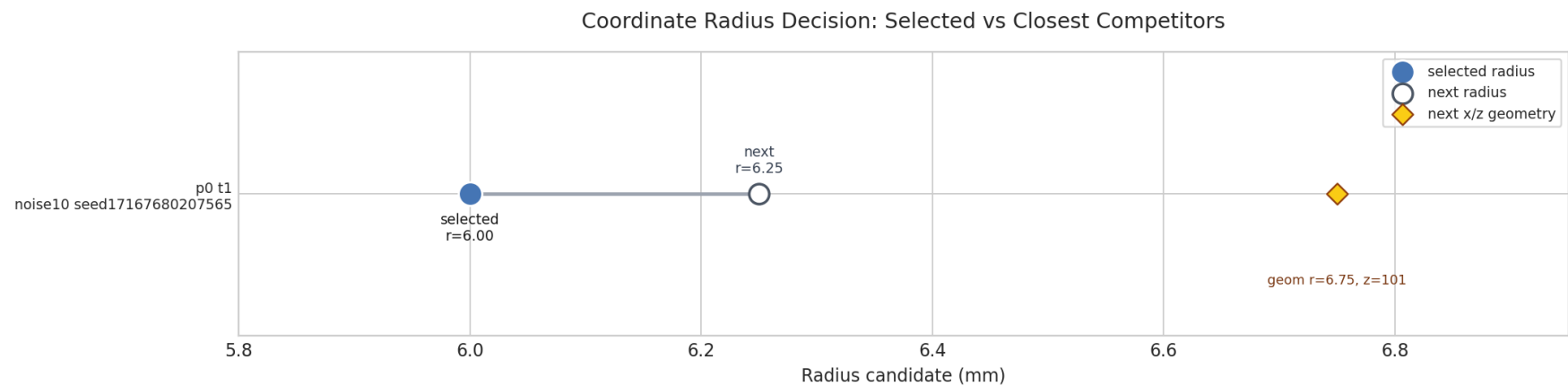


coordinate_optimizer_variable_depth_radius_seed17167680207565_target1_sources9_txrx60_ringdown050_objectives
schematic time 0.00 ns | blue=forward, orange=reflection

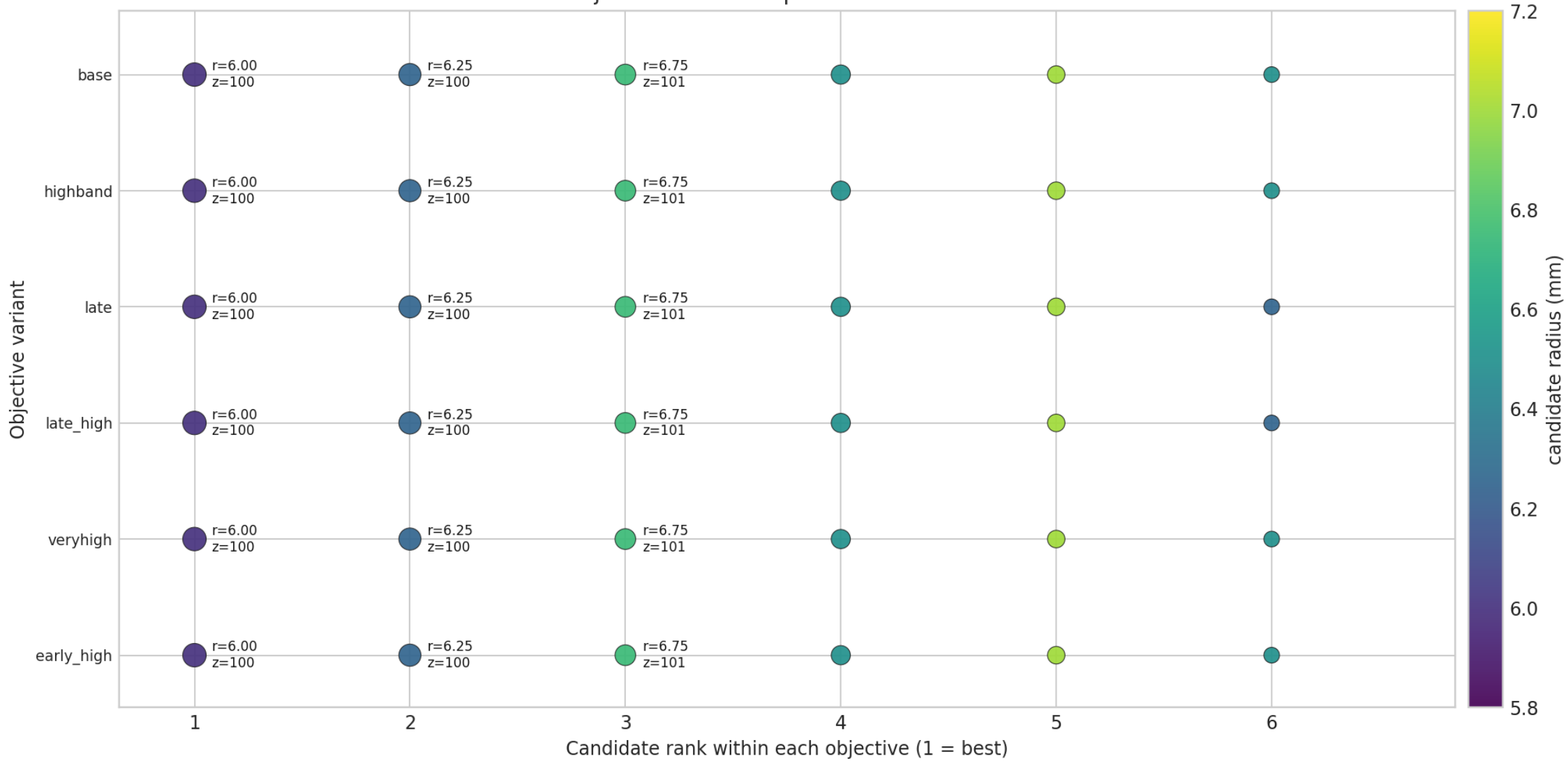


FDTD Ez wavefield | run 1160 target 1 | Tx 194 mm, Rx 254 mm at $t = 0.00$ ns (step 0)





Objective-Variant Top Radius Candidates

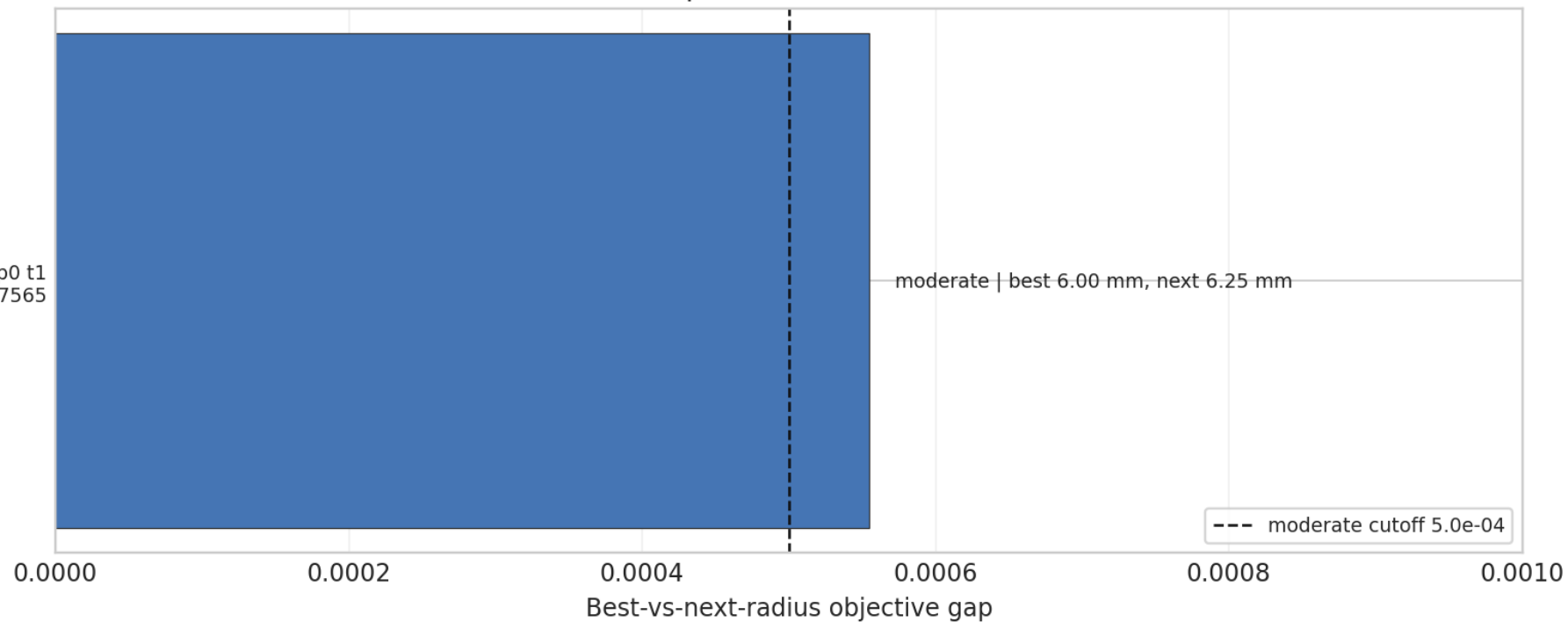


Coordinate Optimizer Radius Confidence

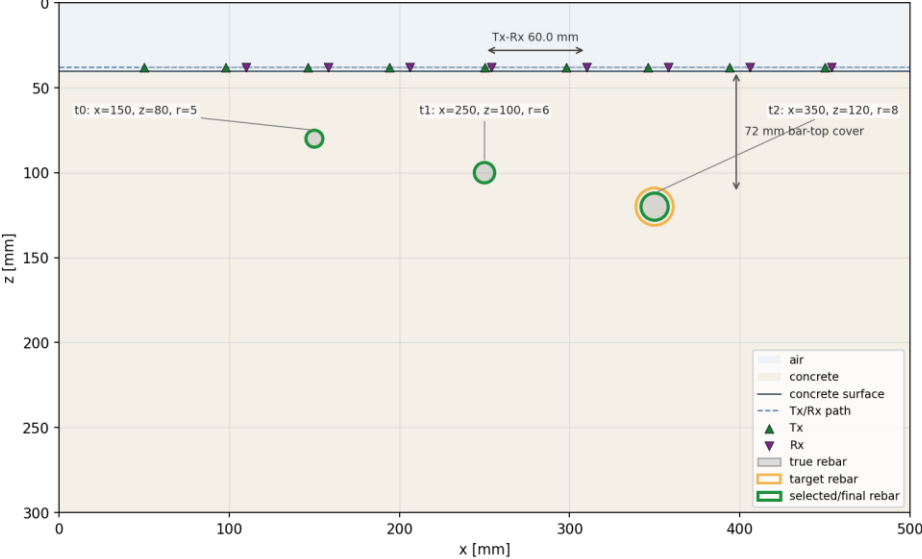
p0 t1
noise10 seed17167680207565

moderate | best 6.00 mm, next 6.25 mm

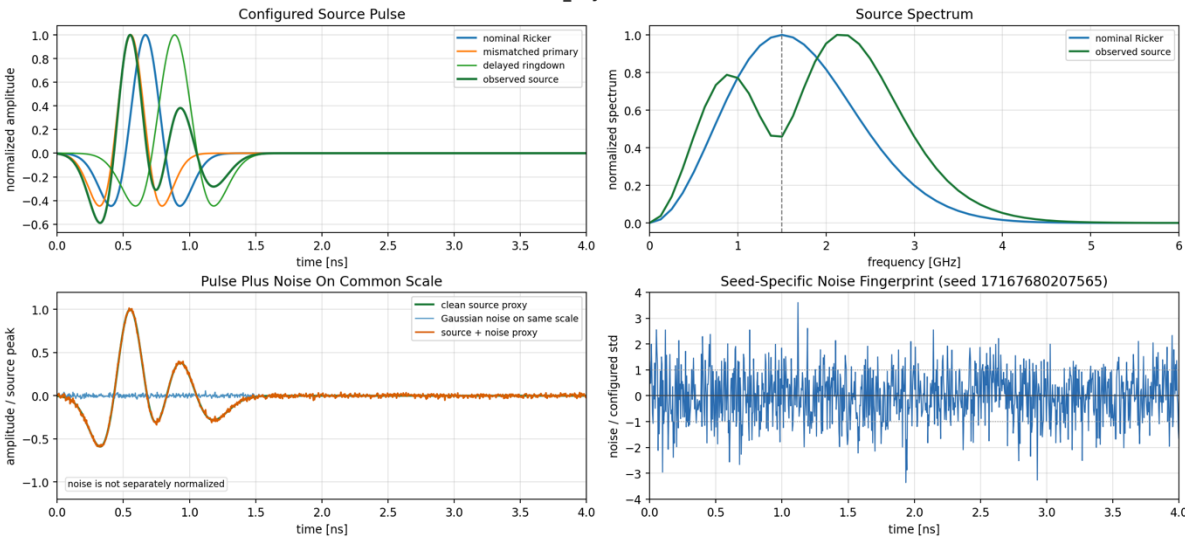
--- moderate cutoff 5.0e-04



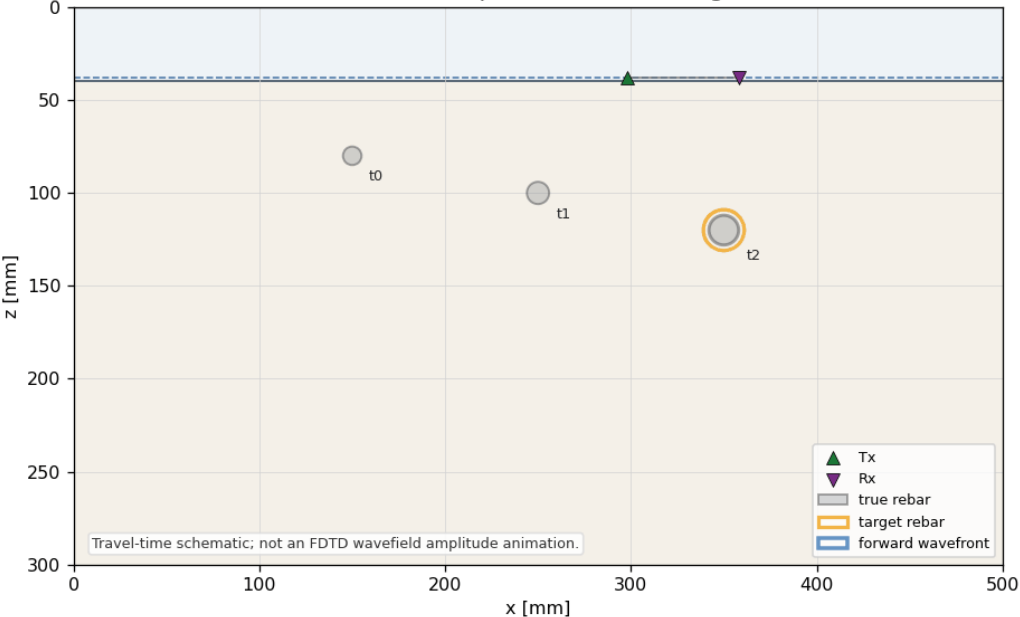
coordinate_optimizer_variable_depth_radius_seed17167680207565_target2_sources9_txrx60_ringdown050_objectives_scene



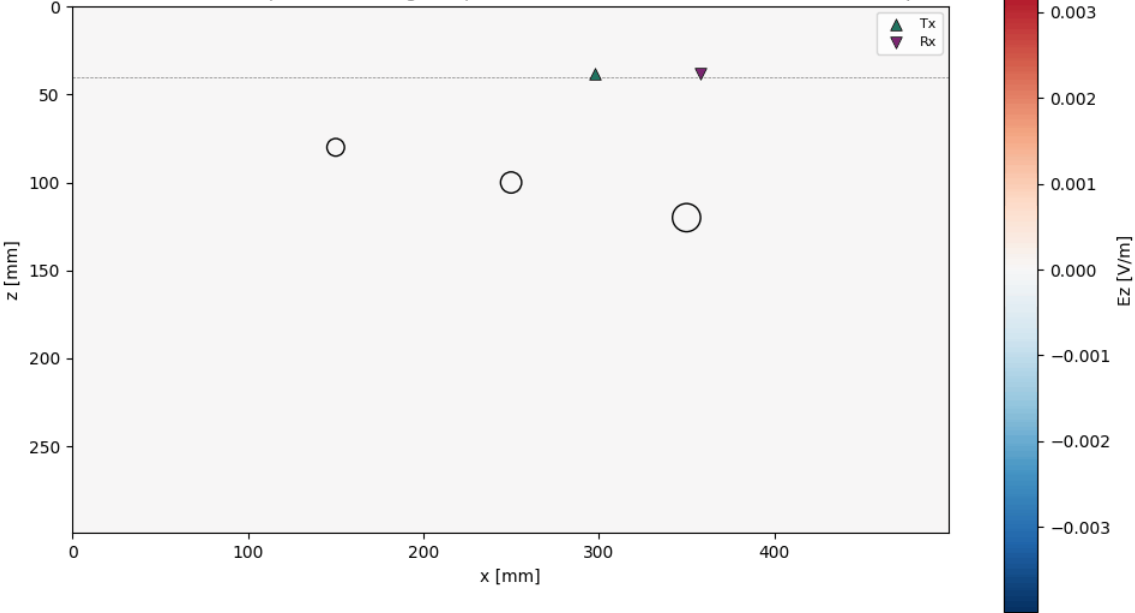
coordinate_optimizer_variable_depth_radius_seed17167680207565_target2_sources9_txrx60_ringdown050_objectives

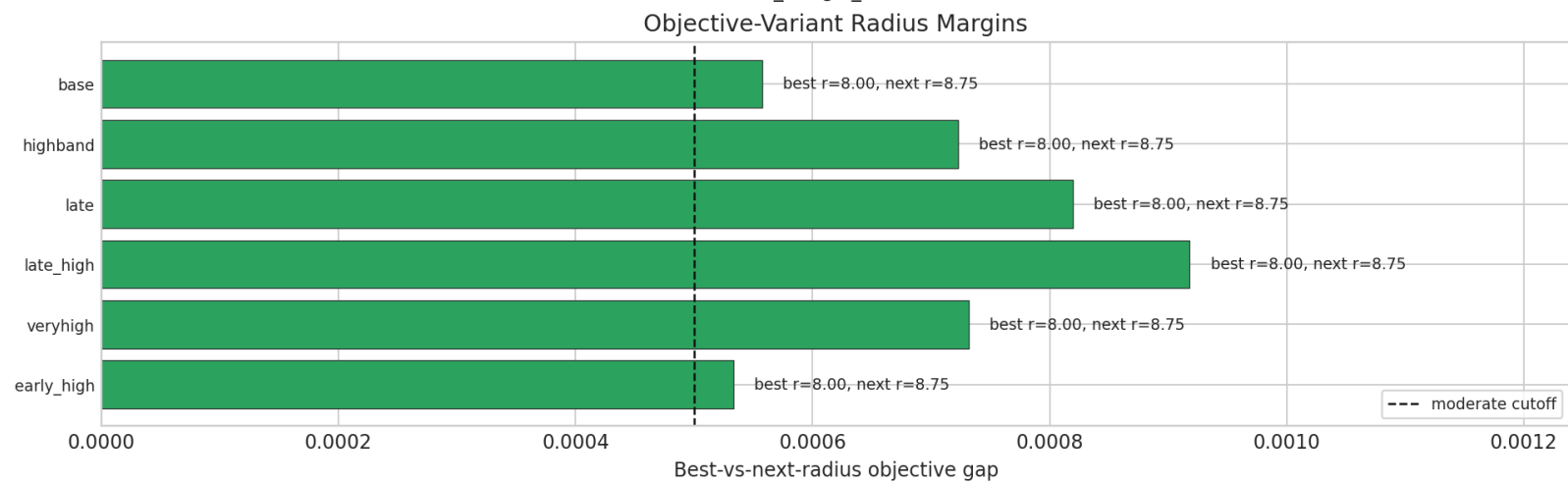
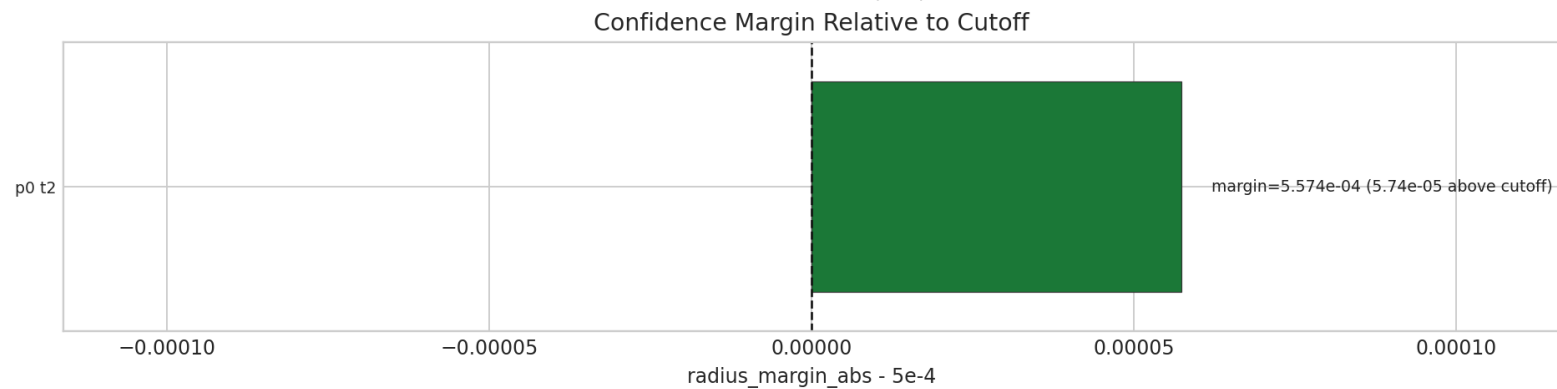
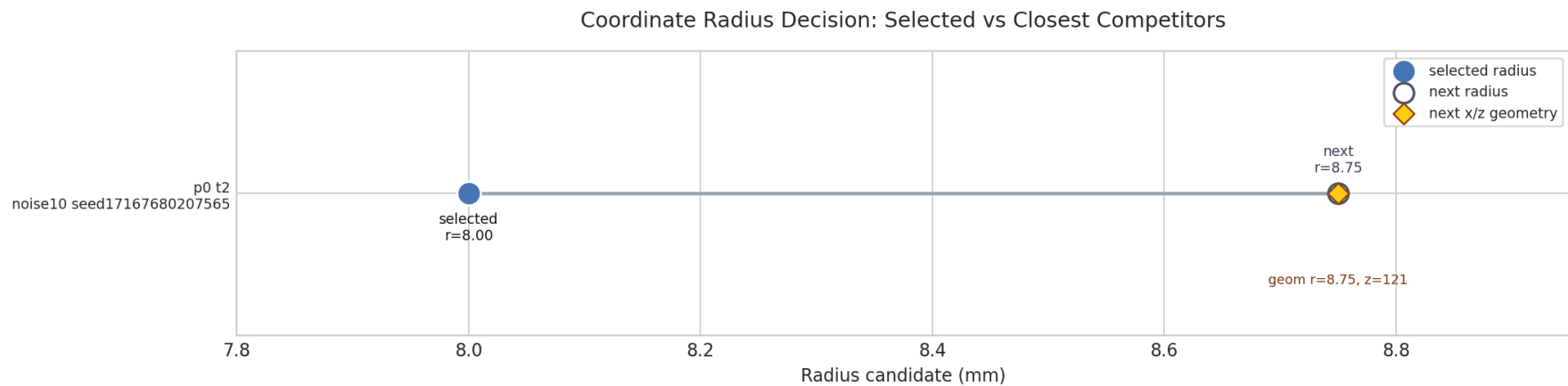


coordinate_optimizer_variable_depth_radius_seed17167680207565_target2_sources9_txrx60_ringdown050_objectives

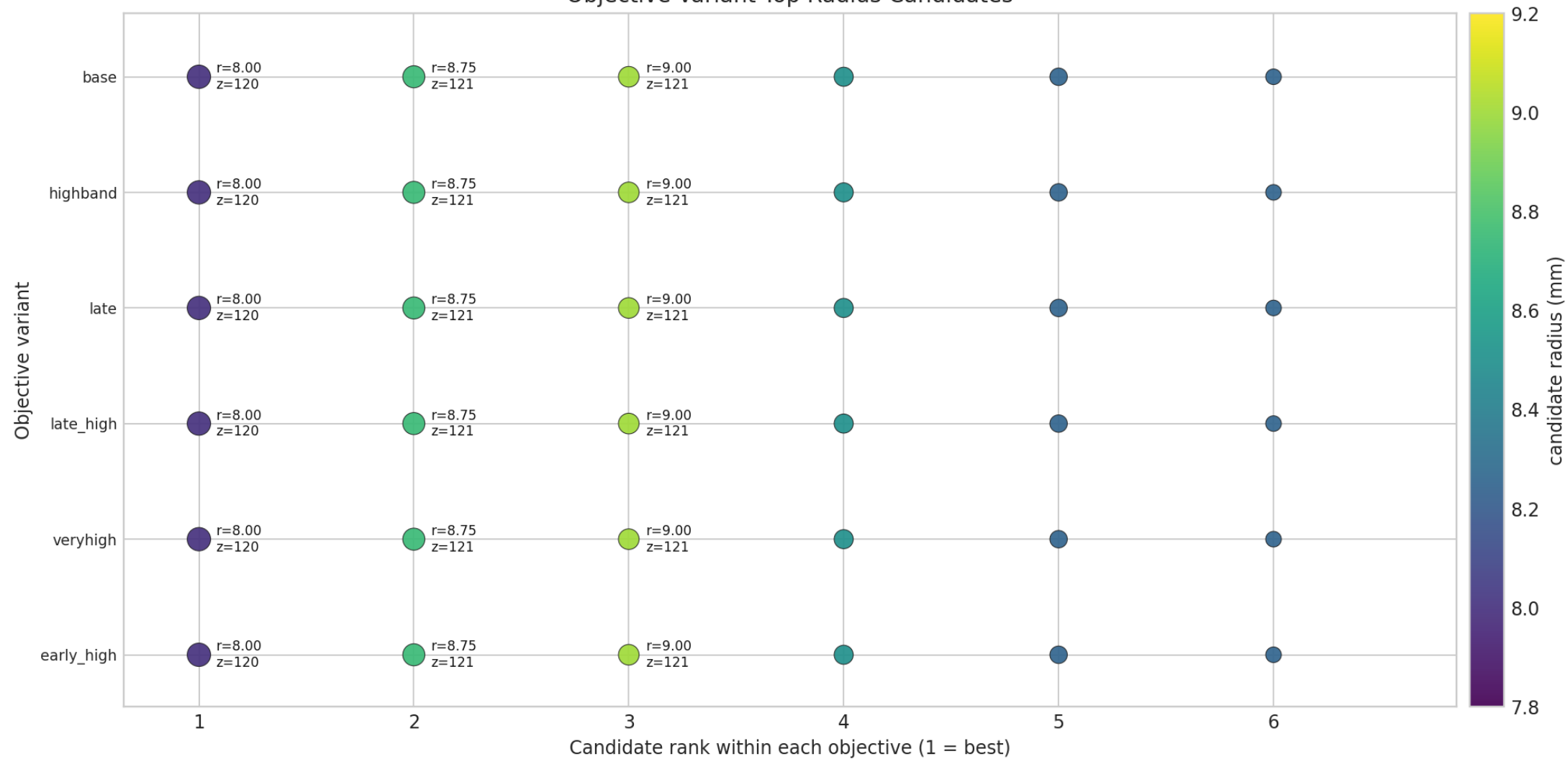


FDTD Ez wavefield | run 1158 target 2 | Tx 298 mm, Rx 358 mm at t = 0.00 ns (step 0)





Objective-Variant Top Radius Candidates

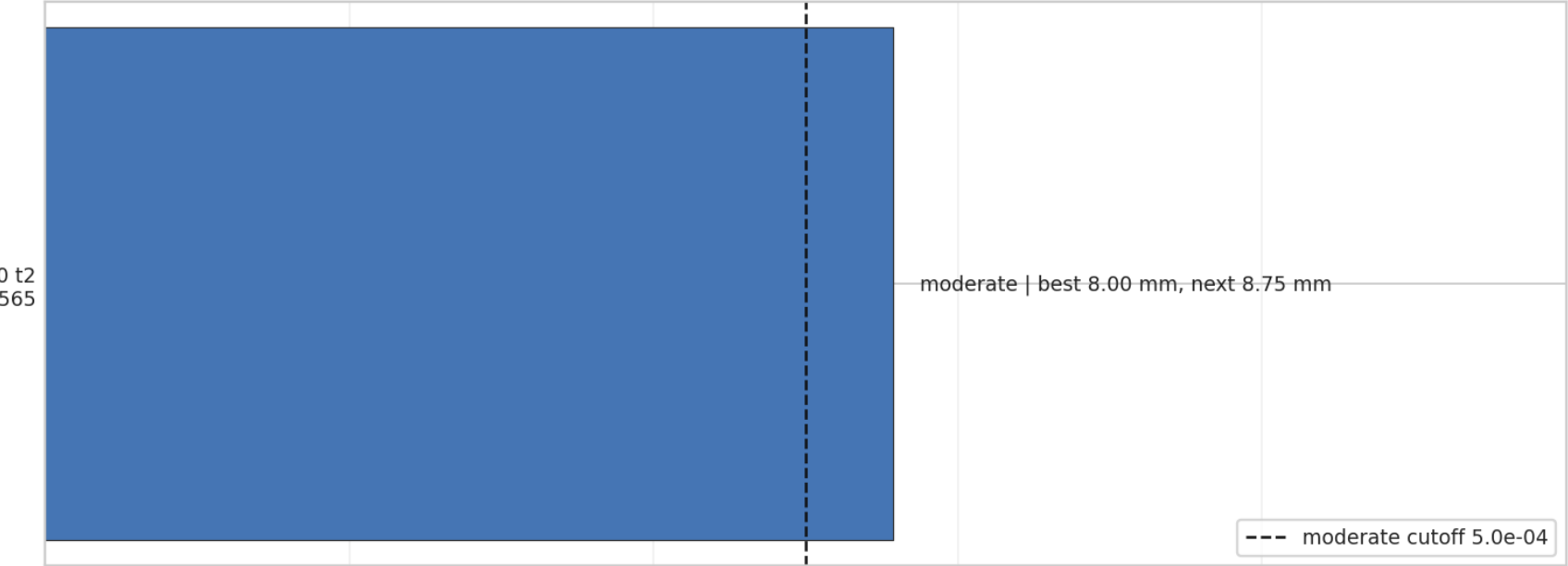


Coordinate Optimizer Radius Confidence

p0 t2
noise10 seed17167680207565

moderate | best 8.00 mm, next 8.75 mm

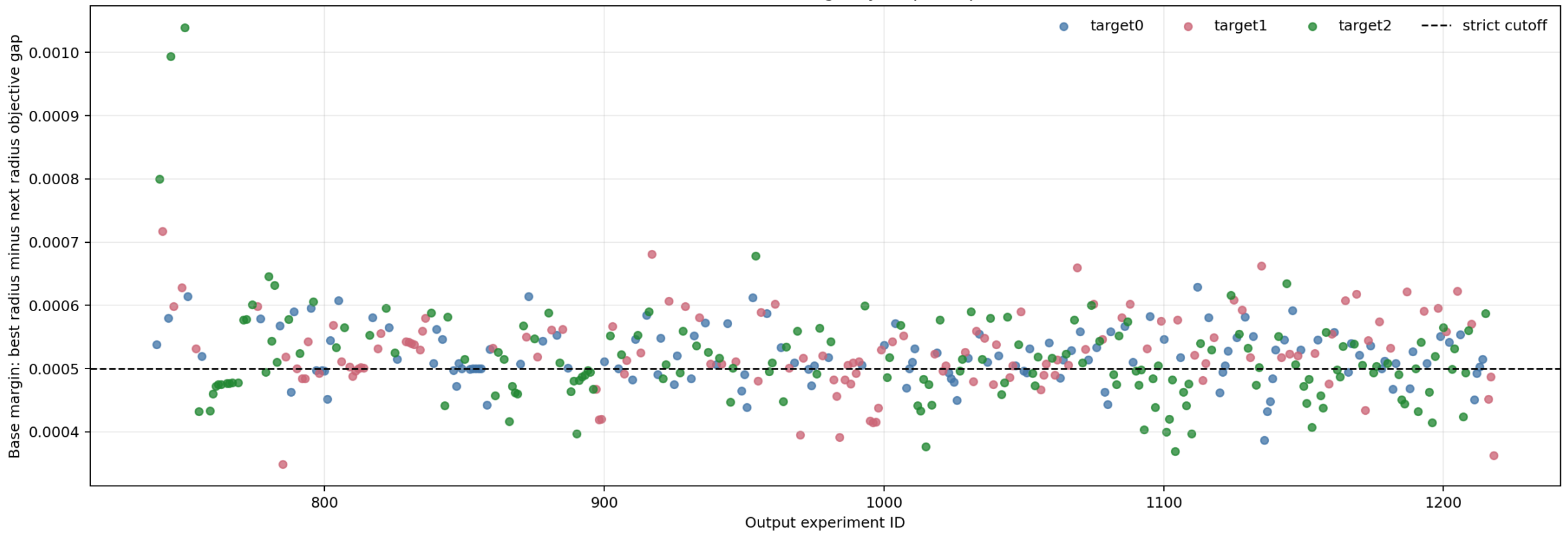
--- moderate cutoff 5.0e-04



0.0000 0.0002 0.0004 0.0006 0.0008 0.0010

Best-vs-next-radius objective gap

Base radius-confidence margin by output experiment ID



Tx/Rx-offset evidence by target

